

V4 parameter estimation

Seven models · SPY, AAPL, MSFT, AMZN · four windows · 1.5-year monthly calibration

What this report is

- The rolling parameter paths used in this study, summarized by regime. One table per model.
- No LSM, no option prices, and no simulated paths. Only calibrate_ticker with the 18-month lookback.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

How estimates are formed

- At monthly update t , parameters use only returns and listed quotes with `trading_date ≤ t``. LSM prices a Monday contract with the latest calibration row dated on or before that Monday. Option quotes are as-of the grid timestamp (same session, else the prior session).
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- Cells are Mean, SD, Min, and Max of those monthly estimates inside the regime.

Tables

- P1 GBM. P2 MD-GBM. P3 GARCH. P4 Heston. P5 Merton. P6 GARCH-Merton. P7 Heston-Merton.
- R1 Crisis, R2 Normal, R3 Late-cycle, R4 COVID. Columns are SPY, AAPL, MSFT, AMZN.

Table P1 · GBM · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

Regime / Parameter	SPY				AAPL				MSFT				AMZN			
Regime / Parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ (P)	-0.2873	0.1250	-0.4424	-0.0608	-0.0499	0.2597	-0.2648	0.4609	-0.2249	0.1181	-0.3572	-0.0067	0.0101	0.2638	-0.3265	0.4808
σ	0.3262	0.0777	0.1852	0.3895	0.5106	0.0533	0.4040	0.5490	0.4089	0.0803	0.2730	0.4812	0.5900	0.0445	0.5045	0.6322
R2 Normal																
μ (P)	0.1779	0.0221	0.1476	0.2233	0.1615	0.2230	-0.1422	0.4632	0.2651	0.0662	0.1597	0.3537	0.1805	0.0801	0.0735	0.3665
σ	0.1120	0.0031	0.1075	0.1165	0.2615	0.0317	0.2210	0.2953	0.2336	0.0039	0.2276	0.2404	0.2943	0.0123	0.2668	0.3063
R3 Late-cycle																
μ (P)	0.0701	0.0323	0.0222	0.1419	0.1466	0.0860	0.0279	0.3211	0.3024	0.0364	0.2586	0.3944	0.3128	0.1220	0.1069	0.5438
σ	0.1452	0.0153	0.1100	0.1582	0.2689	0.0286	0.2073	0.2922	0.2477	0.0186	0.2007	0.2629	0.3172	0.0283	0.2412	0.3409
R4 COVID																
μ (P)	0.0766	0.0617	-0.0804	0.1512	0.3347	0.2027	0.0935	0.7360	0.3438	0.0843	0.2293	0.4846	0.1813	0.1760	-0.0395	0.4934
σ	0.2108	0.0695	0.1452	0.2866	0.3314	0.0460	0.2857	0.3935	0.2962	0.0596	0.2366	0.3640	0.3158	0.0141	0.2976	0.3479

Notes: This table reports Mean, SD, Min, and Max of the monthly GBM estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), σ . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P2 · MD-GBM · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n=18)																
P(U U)	0.4394	0.0177	0.4176	0.4707	0.4833	0.0208	0.4605	0.5123	0.4593	0.0227	0.4315	0.5027	0.4115	0.0124	0.3866	0.4335
P(D D)	0.4131	0.0148	0.3835	0.4312	0.4394	0.0232	0.3899	0.4601	0.5237	0.0105	0.5077	0.5368	0.4731	0.0172	0.4395	0.4951
P(U D)	0.5869	0.0148	0.5688	0.6165	0.5606	0.0232	0.5399	0.6101	0.4763	0.0105	0.4632	0.4923	0.5269	0.0172	0.5049	0.5605
P(D U)	0.5606	0.0177	0.5293	0.5824	0.5167	0.0208	0.4877	0.5395	0.5407	0.0227	0.4973	0.5685	0.5885	0.0124	0.5665	0.6134
μ_U	0.0061	0.0032	0.0000	0.0098	0.0200	0.0014	0.0173	0.0215	0.0131	0.0029	0.0083	0.0164	0.0181	0.0092	0.0000	0.0260
σ_U	0.0143	0.0039	0.0073	0.0172	0.0193	0.0025	0.0147	0.0213	0.0192	0.0042	0.0118	0.0226	0.0286	0.0010	0.0275	0.0311
μ_D	0.0111	0.0026	0.0077	0.0146	0.0201	0.0027	0.0151	0.0229	0.0141	0.0024	0.0109	0.0167	0.0202	0.0027	0.0157	0.0233
σ_D	0.0156	0.0037	0.0082	0.0183	0.0234	0.0023	0.0182	0.0251	0.0173	0.0035	0.0111	0.0206	0.0231	0.0032	0.0168	0.0256
R2 Normal (n=18)																
P(U U)	0.5706	0.0068	0.5588	0.5841	0.4959	0.0354	0.4514	0.5361	0.5133	0.0111	0.4921	0.5270	0.5041	0.0102	0.4894	0.5256
P(D D)	0.4138	0.0186	0.3854	0.4488	0.4606	0.0173	0.4357	0.4810	0.4482	0.0130	0.4331	0.4724	0.4704	0.0185	0.4423	0.4947
P(U D)	0.5862	0.0186	0.5512	0.6146	0.5394	0.0173	0.5190	0.5643	0.5518	0.0130	0.5276	0.5669	0.5296	0.0185	0.5053	0.5577
P(D U)	0.4294	0.0068	0.4159	0.4412	0.5041	0.0354	0.4639	0.5486	0.4867	0.0111	0.4730	0.5079	0.4959	0.0102	0.4744	0.5106
μ_U	0.0047	1.688e-04	0.0043	0.0049	0.0100	0.0011	0.0085	0.0116	0.0085	2.203e-04	0.0082	0.0089	0.0113	4.772e-04	0.0105	0.0119
σ_U	0.0042	2.958e-04	0.0038	0.0047	0.0106	5.961e-04	0.0099	0.0119	0.0099	2.375e-04	0.0095	0.0102	0.0116	3.218e-04	0.0112	0.0125
μ_D	0.0045	1.707e-04	0.0042	0.0048	0.0062	0.0016	0.0026	0.0083	0.0039	0.0011	0.0020	0.0053	0.0054	0.0023	0.0014	0.0092
σ_D	0.0051	1.210e-04	0.0049	0.0053	0.0126	0.0022	0.0099	0.0149	0.0116	1.343e-04	0.0114	0.0118	0.0147	0.0020	0.0100	0.0166
R3 Late-cy (n=18)																
P(U U)	0.5616	0.0074	0.5505	0.5724	0.5468	0.0109	0.5251	0.5613	0.5147	0.0103	0.5000	0.5326	0.5949	0.0133	0.5697	0.6149
P(D D)	0.4588	0.0250	0.4018	0.4942	0.4791	0.0106	0.4622	0.4944	0.3564	0.0093	0.3406	0.3719	0.4643	0.0132	0.4391	0.4879
P(U D)	0.5412	0.0250	0.5058	0.5982	0.5209	0.0106	0.5056	0.5378	0.6436	0.0093	0.6281	0.6594	0.5357	0.0132	0.5121	0.5609
P(D U)	0.4384	0.0074	0.4276	0.4495	0.4532	0.0109	0.4387	0.4749	0.4853	0.0103	0.4674	0.5000	0.4051	0.0133	0.3851	0.4303
μ_U	0.0044	6.179e-04	0.0036	0.0052	0.0090	6.684e-04	0.0079	0.0101	0.0080	8.168e-04	0.0069	0.0092	0.0089	0.0011	0.0076	0.0106
σ_U	0.0057	6.881e-04	0.0042	0.0062	0.0112	9.124e-04	0.0095	0.0120	0.0107	7.615e-04	0.0090	0.0115	0.0138	0.0012	0.0112	0.0153
μ_D	0.0024	0.0017	0.0000	0.0045	0.0078	5.336e-04	0.0067	0.0085	0.0083	0.0014	0.0056	0.0099	0.0097	0.0016	0.0066	0.0115
σ_D	0.0077	7.488e-04	0.0059	0.0082	0.0130	0.0020	0.0086	0.0147	0.0115	8.387e-04	0.0093	0.0122	0.0153	0.0016	0.0106	0.0165
R4 COVID (n=18)																
P(U U)	0.5619	0.0170	0.5390	0.5920	0.5440	0.0200	0.5197	0.5764	0.5435	0.0157	0.5186	0.5691	0.5445	0.0213	0.5026	0.5738
P(D D)	0.4425	0.0435	0.3703	0.4856	0.4342	0.0323	0.3811	0.4829	0.3817	0.0219	0.3494	0.4156	0.4673	0.0120	0.4494	0.4918
P(U D)	0.5575	0.0435	0.5144	0.6297	0.5658	0.0323	0.5171	0.6189	0.6183	0.0219	0.5844	0.6506	0.5327	0.0120	0.5082	0.5506
P(D U)	0.4381	0.0170	0.4080	0.4610	0.4560	0.0200	0.4236	0.4803	0.4565	0.0157	0.4309	0.4814	0.4555	0.0213	0.4262	0.4974
μ_U	0.0026	0.0025	0.0000	0.0051	0.0104	5.222e-04	0.0097	0.0115	0.0069	0.0021	0.0037	0.0092	0.0092	8.631e-04	0.0080	0.0106
σ_U	0.0084	0.0028	0.0058	0.0115	0.0137	0.0027	0.0108	0.0169	0.0129	0.0034	0.0097	0.0168	0.0139	8.781e-04	0.0130	0.0157
μ_D	0.0021	0.0020	0.0000	0.0045	0.0049	0.0036	0.0000	0.0087	0.0052	0.0042	0.0000	0.0096	0.0096	7.556e-04	0.0087	0.0113
σ_D	0.0107	0.0033	0.0075	0.0143	0.0170	0.0025	0.0146	0.0206	0.0148	0.0039	0.0110	0.0192	0.0148	8.454e-04	0.0133	0.0163

Notes: This table reports Mean, SD, Min, and Max of the monthly MD-GBM estimates for SPY, AAPL, MSFT, AMZN. Parameters are P(U|U), P(D|D), P(U|D), P(D|U), μ_U, σ_U, μ_D, σ_D. Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P3 · GARCH · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

Regime / para	SPY				AAPL				MSFT				AMZN			
Regime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n=18)																
λ	-0.0247	0.0178	-0.0470	0.0104	0.0718	0.0302	0.0442	0.1274	-0.0162	0.0123	-0.0340	0.0058	0.0415	0.0234	0.0056	0.0800
ω	5.610e-06	1.728e-06	2.700e-06	8.327e-06	4.785e-05	1.437e-05	1.535e-05	6.194e-05	2.274e-05	8.998e-06	1.196e-05	4.171e-05	4.422e-04	4.626e-04	1.881e-05	9.889e-04
α	0.1113	0.0216	0.0553	0.1328	0.1226	0.0166	0.0991	0.1532	0.1617	0.0431	0.1023	0.2160	0.1583	0.1450	0.0461	0.4570
β	0.8787	0.0192	0.8539	0.9276	0.8334	0.0260	0.7917	0.8766	0.8202	0.0356	0.7726	0.8845	0.4938	0.4761	7.032e-54	0.9354
σ_0	0.0254	0.0127	0.0125	0.0501	0.0344	0.0191	0.0177	0.0867	0.0325	0.0126	0.0151	0.0549	0.0351	0.0083	0.0262	0.0583
persist P	0.9900	0.0059	0.9799	1.0000	0.9560	0.0199	0.9123	0.9878	0.9819	0.0134	0.9578	1.0000	0.6521	0.3806	0.1091	0.9865
persist Q	0.9901	0.0059	0.9801	1.0000	0.9567	0.0197	0.9142	0.9890	0.9820	0.0134	0.9578	1.0002	0.6524	0.3803	0.1097	0.9866
R2 Normal (n=18)																
λ	0.1362	0.0122	0.1186	0.1614	0.0647	0.0534	-0.0073	0.1415	0.1320	0.0461	0.0695	0.1863	0.0372	0.0308	0.0108	0.1122
ω	1.090e-05	3.206e-06	5.846e-06	1.521e-05	2.536e-06	4.270e-06	8.356e-43	1.192e-05	1.104e-04	3.669e-05	7.090e-05	1.662e-04	2.147e-04	8.497e-05	4.034e-05	2.967e-04
α	0.1600	0.0339	0.1222	0.2220	0.0083	0.0139	1.645e-129	0.0381	0.3529	0.1915	0.1078	0.6187	0.3203	0.1088	0.1213	0.4827
β	0.6275	0.0726	0.5147	0.7307	0.9842	0.0245	0.9365	0.9997	0.2580	0.0593	0.1836	0.3705	0.1081	0.2642	7.014e-15	0.7372
σ_0	0.0064	9.323e-04	0.0049	0.0083	0.0147	0.0025	0.0126	0.0216	0.0136	0.0016	0.0107	0.0167	0.0173	0.0038	0.0142	0.0291
persist P	0.7875	0.0555	0.7088	0.8780	0.9924	0.0113	0.9691	0.9997	0.6110	0.2341	0.2914	0.8652	0.4284	0.2185	0.2281	0.9031
persist Q	0.7906	0.0553	0.7108	0.8813	0.9924	0.0113	0.9691	0.9997	0.6197	0.2413	0.2921	0.8843	0.4289	0.2190	0.2282	0.9051
R3 Late-cy (n=18)																
λ	0.1459	0.0280	0.1055	0.1865	0.1067	0.0194	0.0746	0.1370	0.1382	0.0150	0.1141	0.1720	0.1559	0.0249	0.1017	0.1899
ω	3.612e-06	1.256e-06	1.959e-06	5.785e-06	2.245e-05	9.679e-06	1.083e-05	4.363e-05	2.908e-05	9.041e-06	1.511e-05	4.244e-05	3.489e-05	1.797e-05	1.360e-05	6.055e-05
α	0.2097	0.0272	0.1665	0.2516	0.1526	0.0239	0.1146	0.1902	0.2085	0.0294	0.1355	0.2601	0.1956	0.0328	0.1293	0.2527
β	0.7631	0.0213	0.7216	0.7965	0.7692	0.0634	0.6206	0.8568	0.6706	0.0717	0.5159	0.7981	0.7202	0.0492	0.6160	0.8079
σ_0	0.0095	0.0052	0.0044	0.0230	0.0169	0.0060	0.0108	0.0288	0.0154	0.0051	0.0096	0.0269	0.0203	0.0097	0.0125	0.0413
persist P	0.9728	0.0226	0.9381	1.0000	0.9218	0.0734	0.7440	0.9714	0.8791	0.0664	0.7167	0.9470	0.9158	0.0688	0.7453	0.9796
persist Q	0.9775	0.0239	0.9443	1.0081	0.9236	0.0736	0.7458	0.9723	0.8831	0.0658	0.7226	0.9527	0.9207	0.0686	0.7500	0.9863
R4 COVID (n=18)																
λ	0.1571	0.0275	0.1093	0.1929	0.1688	0.0400	0.1151	0.2350	0.1396	0.0186	0.1141	0.1685	0.0929	0.0231	0.0502	0.1309
ω	4.488e-06	5.784e-07	3.298e-06	5.817e-06	1.705e-05	2.450e-06	1.309e-05	2.072e-05	1.271e-05	3.167e-06	7.421e-06	2.029e-05	1.614e-05	5.812e-06	1.074e-05	2.975e-05
α	0.2515	0.0569	0.1677	0.3259	0.1616	0.0301	0.1224	0.2273	0.1858	0.0367	0.1355	0.2400	0.1510	0.0324	0.1034	0.2019
β	0.7293	0.0386	0.6731	0.7890	0.8052	0.0268	0.7506	0.8388	0.7736	0.0305	0.7227	0.8060	0.8081	0.0443	0.7164	0.8659
σ_0	0.0142	0.0127	0.0053	0.0487	0.0210	0.0116	0.0121	0.0536	0.0201	0.0144	0.0092	0.0597	0.0177	0.0062	0.0119	0.0333
persist P	0.9808	0.0211	0.9467	1.0000	0.9668	0.0120	0.9453	0.9912	0.9595	0.0201	0.9292	0.9943	0.9591	0.0176	0.9156	0.9737
persist Q	0.9875	0.0240	0.9489	1.0115	0.9719	0.0127	0.9528	0.9944	0.9633	0.0211	0.9318	0.9977	0.9605	0.0172	0.9180	0.9748

Notes: This table reports Mean, SD, Min, and Max of the monthly GARCH estimates for SPY, AAPL, MSFT, AMZN. Parameters are λ , ω , α , β , σ_0 , persist P, persist Q. Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P4 · Heston · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

regime / para	SPY				AAPL				MSFT				AMZN			
regime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ (P)	-0.2873	0.1250	-0.4424	-0.0608	-0.0499	0.2597	-0.2648	0.4609	-0.2249	0.1181	-0.3572	-0.0067	0.0101	0.2638	-0.3265	0.4808
κ (P)	18.6154	4.9923	2.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000
θ (P)	0.1121	0.0453	0.0343	0.1517	0.2629	0.0507	0.1636	0.3008	0.1730	0.0609	0.0744	0.2311	0.3492	0.0503	0.2547	0.3987
ξ	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000
ρ	-0.1421	0.0901	-0.4293	-0.0739	-0.2095	0.0761	-0.3689	-0.0647	0.1209	0.0862	-0.0854	0.2596	0.1662	0.1090	0.0665	0.3595
v_0 (P)	0.1825	0.1861	0.0337	0.6642	0.3009	0.2859	0.0520	0.8809	0.2744	0.2284	0.0534	0.8709	0.4553	0.3200	0.0900	1.0091
η_v (prem)	-5.5587	20.0531	-40.0000	11.5216	-0.5564	11.5846	-26.8401	12.5193	-4.9497	17.9213	-39.5028	15.5285	-1.1411	12.4099	-29.0924	15.1957
κ^* (Q)	24.1741	21.2079	-0.0163	60.0000	20.5564	11.5846	7.4807	46.8401	24.9497	17.9213	4.4715	59.5028	21.1411	12.4099	4.8043	49.0924
θ^* (Q)	-0.2135	1.2401	-4.3338	0.2462	0.3332	0.1918	0.1142	0.7635	0.2179	0.1585	0.0737	0.5455	0.4860	0.3917	0.1552	1.4799
$v_0^{\wedge}Q$	0.1209	0.0993	0.0234	0.3625	0.2341	0.1863	0.0133	0.5740	0.1475	0.0784	0.0596	0.3521	0.3971	0.3861	1.000e-04	1.1427
n quotes	23.4615	0.7763	22.0000	24.0000	9.6923	2.2871	6.0000	14.0000	9.3077	4.6795	6.0000	24.0000	7.6923	4.9562	6.0000	24.0000
R2 Normal																
μ (P)	0.1779	0.0221	0.1476	0.2233	0.1615	0.2230	-0.1422	0.4632	0.2651	0.0662	0.1597	0.3537	0.1805	0.0801	0.0735	0.3665
κ (P)	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000
θ (P)	0.0127	6.929e-04	0.0117	0.0137	0.0694	0.0162	0.0494	0.0871	0.0547	0.0018	0.0519	0.0579	0.0867	0.0070	0.0715	0.0936
ξ	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000
ρ	-0.1107	0.0640	-0.2083	0.0165	-0.1122	0.1124	-0.2339	0.0121	-0.1845	0.0086	-0.2005	-0.1724	-0.0713	0.1144	-0.1768	0.2440
v_0 (P)	0.0127	0.0083	0.0034	0.0331	0.0456	0.0357	0.0115	0.1235	0.0364	0.0177	0.0150	0.0697	0.1146	0.0863	0.0309	0.2513
η_v (prem)	-32.0660	19.3753	-40.0000	12.9851	-1.2188	14.9489	-24.1003	19.9805	-11.1278	10.6332	-31.0505	6.8401	0.0096	15.8628	-39.3942	19.6899
κ^* (Q)	52.0660	19.3753	7.0149	60.0000	21.2188	14.9489	0.0195	44.1003	31.1278	10.6332	13.1599	51.0505	19.9904	15.8628	0.3101	59.3942
θ^* (Q)	0.0081	0.0098	0.0039	0.0352	4.1318	14.5418	0.0351	52.5289	0.0397	0.0161	0.0206	0.0831	0.7388	1.6783	0.0262	6.0349
$v_0^{\wedge}Q$	0.0356	0.0147	0.0120	0.0634	0.0664	0.0855	0.0125	0.3431	0.0484	0.0177	0.0223	0.0838	0.1523	0.2499	0.0484	0.9735
n quotes	20.3846	2.9308	15.0000	24.0000	23.6923	0.6304	22.0000	24.0000	18.8462	2.8239	13.0000	24.0000	23.8462	0.3755	23.0000	24.0000
R3 Late-cy																
μ (P)	0.0701	0.0323	0.0222	0.1419	0.1466	0.0860	0.0279	0.3211	0.3024	0.0364	0.2586	0.3944	0.3128	0.1220	0.1069	0.5438
κ (P)	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000
θ (P)	0.0213	0.0041	0.0121	0.0250	0.0730	0.0143	0.0433	0.0852	0.0619	0.0087	0.0408	0.0693	0.1015	0.0164	0.0592	0.1166
ξ	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000
ρ	-0.2517	0.0784	-0.4064	-0.1552	-0.0895	0.1186	-0.1953	0.1557	0.0480	0.0609	-0.0848	0.1792	0.1043	0.0758	0.0180	0.2105
v_0 (P)	0.0264	0.0248	0.0032	0.0848	0.1006	0.0733	0.0215	0.2477	0.0694	0.0532	0.0210	0.1819	0.1213	0.1161	0.0252	0.3717
η_v (prem)	-24.5886	21.0171	-40.0000	19.3512	-5.8006	18.8634	-40.0000	17.8335	-10.7461	20.7475	-40.0000	11.7931	-4.7623	16.8084	-40.0000	19.3680
κ^* (Q)	44.5886	21.0171	0.6488	60.0000	25.8006	18.8634	2.1665	60.0000	30.7461	20.7475	8.2069	60.0000	24.7623	16.8084	0.6320	60.0000
θ^* (Q)	0.0603	0.1758	0.0049	0.6446	0.1093	0.1046	0.0282	0.3993	0.0624	0.0420	0.0219	0.1451	0.2416	0.4943	0.0365	1.8740
$v_0^{\wedge}Q$	0.0329	0.0161	0.0124	0.0615	0.1395	0.0979	0.0132	0.2929	0.0712	0.0726	0.0086	0.2458	0.1278	0.1350	0.0106	0.3920
n quotes	23.9231	0.2774	23.0000	24.0000	23.9231	0.2774	23.0000	24.0000	21.8462	2.5445	14.0000	24.0000	24.0000	0.0000	24.0000	24.0000
R4 COVID																
μ (P)	0.0766	0.0617	-0.0804	0.1512	0.3347	0.2027	0.0935	0.7360	0.3438	0.0843	0.2293	0.4846	0.1813	0.1760	-0.0395	0.4934
κ (P)	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000	20.0000	0.0000	20.0000	20.0000
θ (P)	0.0488	0.0300	0.0211	0.0820	0.1121	0.0311	0.0817	0.1546	0.0913	0.0359	0.0563	0.1327	0.0999	0.0090	0.0890	0.1207
ξ	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000	3.0000	0.0000	3.0000	3.0000
ρ	-0.2295	0.0689	-0.3397	-0.1561	-0.1575	0.0735	-0.2458	-0.0105	-0.0838	0.0464	-0.1799	0.0359	0.0315	0.0490	-0.0910	0.0972
v_0 (P)	0.1050	0.2354	0.0038	0.8731	0.1685	0.2515	0.0306	0.9779	0.1597	0.3094	0.0084	1.1708	0.1138	0.1090	0.0159	0.4135
η_v (prem)	-23.3772	18.8736	-40.0000	4.7721	-6.9393	13.2095	-40.0000	7.9088	-8.5170	15.0237	-35.0248	11.7931	-7.8275	15.5904	-35.3364	12.8521
κ^* (Q)	43.3772	18.8736	15.2279	60.0000	26.9393	13.2095	12.0912	60.0000	28.5170	15.0237	8.2069	55.0248	27.8275	15.5904	7.1479	55.3364
θ^* (Q)	0.0362	0.0375	0.0070	0.1058	0.1005	0.0494	0.0277	0.1941	0.0848	0.0527	0.0207	0.1977	0.1059	0.0745	0.0350	0.2491
$v_0^{\wedge}Q$	0.0710	0.1526	0.0064	0.5735	0.1872	0.1535	0.0309	0.4857	0.1451	0.1810	0.0131	0.6557	0.1597	0.1653	0.0106	0.4867
n quotes	24.0000	0.0000	24.0000	24.0000	24.0000	0.0000	24.0000	24.0000	23.3077	0.9473	21.0000	24.0000	24.0000	0.0000	24.0000	24.0000

Notes: This table reports Mean, SD, Min, and Max of the monthly Heston estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), κ (P), θ (P), ξ , ρ , v_0 (P), η_v (premium), κ^* (Q), θ^* (Q), $v_0^{\wedge}Q$, n quotes. Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P5 · Merton · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ (P)	-0.2184	0.0900	-0.3501	-0.0707	0.0013	0.2783	-0.2261	0.5259	-0.3118	0.2108	-0.5198	0.0731	-0.1561	0.1577	-0.4766	0.0916
σ	0.2709	0.0606	0.1725	0.3288	0.4442	0.0414	0.3678	0.4777	0.3424	0.0678	0.2379	0.4109	0.5005	0.0709	0.3758	0.5664
$\lambda^* = \lambda$	3.8907	1.0064	1.3263	5.3053	2.5088	0.2923	1.9895	2.6880	3.5830	0.5718	2.6526	4.6421	3.8377	0.6586	2.6737	5.2775
μ_J (P)	-0.0146	0.0205	-0.0308	0.0388	-0.0202	0.0200	-0.0750	0.0170	0.0211	0.0269	-0.0259	0.0485	0.0438	0.0351	0.0069	0.1064
σ_J	0.0833	0.0307	0.0036	0.1011	0.1554	0.0222	0.1145	0.1858	0.1130	0.0218	0.0778	0.1328	0.1468	0.0091	0.1304	0.1608
κ (P)	-0.0105	0.0191	-0.0261	0.0396	-0.0076	0.0221	-0.0649	0.0349	0.0284	0.0293	-0.0226	0.0559	0.0568	0.0368	0.0180	0.1218
μ_J^* (Q)	0.0043	0.0751	-0.0821	0.1725	-0.0182	0.1649	-0.3418	0.3017	0.0045	0.0929	-0.1056	0.2046	0.0546	0.1559	-0.1061	0.3501
κ^Q	0.0110	0.0787	-0.0741	0.1931	0.0072	0.1743	-0.2839	0.3707	0.0153	0.0979	-0.0975	0.2348	0.0801	0.1781	-0.0908	0.4355
n quotes	23.4615	0.7763	22.0000	24.0000	9.6923	2.2871	6.0000	14.0000	9.3077	4.6795	6.0000	24.0000	7.6923	4.9562	6.0000	24.0000
R2 Normal																
μ (P)	0.2198	0.0276	0.1728	0.2727	0.2594	0.1258	0.0754	0.4440	0.3161	0.0715	0.2188	0.4153	0.3107	0.0432	0.2152	0.3748
σ	0.1016	0.0037	0.0965	0.1065	0.2081	0.0277	0.1714	0.2411	0.1796	0.0048	0.1723	0.1884	0.2289	0.0067	0.2170	0.2365
$\lambda^* = \lambda$	3.5304	0.6309	1.9791	4.6667	4.2451	0.5021	3.3245	4.6790	3.1212	0.3254	2.6387	3.3511	3.6845	0.3583	3.2984	4.0213
μ_J (P)	-0.0108	0.0043	-0.0153	-0.0049	-0.0245	0.0288	-0.0541	0.0059	-0.0147	0.0034	-0.0169	-0.0098	-0.0364	0.0300	-0.0639	0.0258
σ_J	0.0223	0.0031	0.0182	0.0264	0.0687	0.0050	0.0644	0.0791	0.0835	0.0051	0.0802	0.0908	0.0855	0.0039	0.0728	0.0873
κ (P)	-0.0105	0.0043	-0.0150	-0.0045	-0.0215	0.0280	-0.0505	0.0080	-0.0112	0.0038	-0.0136	-0.0057	-0.0318	0.0291	-0.0584	0.0289
μ_J^* (Q)	-0.0072	0.0082	-0.0234	0.0122	-0.0329	0.0372	-0.1172	0.0198	-0.0274	0.0130	-0.0420	0.0092	-0.0090	0.0865	-0.1779	0.1664
κ^Q	-0.0068	0.0082	-0.0230	0.0126	-0.0295	0.0357	-0.1086	0.0223	-0.0236	0.0126	-0.0372	0.0125	-0.0019	0.0866	-0.1599	0.1856
n quotes	20.3846	2.9308	15.0000	24.0000	23.6923	0.6304	22.0000	24.0000	18.8462	2.8239	13.0000	24.0000	23.8462	0.3755	23.0000	24.0000
R3 Late-cy																
μ (P)	0.1458	0.0444	0.0916	0.2578	0.1514	0.0680	0.0507	0.2592	0.2827	0.0605	0.2005	0.4313	0.2871	0.1181	0.0986	0.5329
σ	0.1201	0.0139	0.0866	0.1321	0.2268	0.0225	0.1723	0.2454	0.2090	0.0204	0.1625	0.2270	0.2656	0.0315	0.1942	0.2915
$\lambda^* = \lambda$	4.6249	0.3420	3.9685	5.3333	4.0056	0.7597	3.3333	6.0000	4.2127	0.6868	2.6667	5.3475	4.2115	0.8252	2.0000	5.3333
μ_J (P)	-0.0159	0.0049	-0.0239	-0.0107	-0.0011	0.0081	-0.0096	0.0164	0.0058	0.0073	-0.0075	0.0170	0.0078	0.0156	-0.0189	0.0337
σ_J	0.0337	0.0060	0.0208	0.0377	0.0727	0.0140	0.0440	0.0836	0.0642	0.0024	0.0585	0.0676	0.0832	0.0064	0.0738	0.1010
κ (P)	-0.0152	0.0049	-0.0233	-0.0100	0.0017	0.0079	-0.0079	0.0175	0.0079	0.0074	-0.0058	0.0194	0.0114	0.0161	-0.0155	0.0381
μ_J^* (Q)	-0.0327	0.0372	-0.1253	-0.0089	0.0141	0.0710	-0.0551	0.1606	-0.0016	0.0493	-0.0511	0.1011	0.0199	0.0824	-0.0500	0.2014
κ^Q	-0.0310	0.0351	-0.1173	-0.0086	0.0194	0.0740	-0.0523	0.1767	0.0016	0.0505	-0.0482	0.1085	0.0269	0.0881	-0.0457	0.2277
n quotes	23.9231	0.2774	23.0000	24.0000	23.9231	0.2774	23.0000	24.0000	21.8462	2.5445	14.0000	24.0000	24.0000	0.0000	24.0000	24.0000
R4 COVID																
μ (P)	0.1067	0.0647	-0.0407	0.1676	0.3576	0.1562	0.1487	0.6455	0.3393	0.0852	0.2152	0.4715	0.1547	0.1443	-0.1547	0.3835
σ	0.1516	0.0309	0.1210	0.1903	0.2613	0.0222	0.2376	0.3003	0.2321	0.0223	0.2072	0.2662	0.2640	0.0198	0.2450	0.3004
$\lambda^* = \lambda$	5.2356	0.6039	4.6421	6.0319	4.6200	0.7491	3.9789	6.0000	4.3121	1.2686	2.6667	6.0000	5.1816	1.2132	3.3511	6.6667
μ_J (P)	-0.0057	0.0078	-0.0172	0.0075	-0.0047	0.0137	-0.0278	0.0252	0.0033	0.0097	-0.0148	0.0170	0.0089	0.0186	-0.0133	0.0426
σ_J	0.0607	0.0285	0.0335	0.0920	0.0922	0.0156	0.0751	0.1118	0.0838	0.0225	0.0615	0.1122	0.0740	0.0032	0.0706	0.0802
κ (P)	-0.0034	0.0094	-0.0165	0.0118	-2.834e-04	0.0143	-0.0247	0.0308	0.0071	0.0087	-0.0099	0.0194	0.0119	0.0189	-0.0105	0.0462
μ_J^* (Q)	-0.0110	0.0692	-0.0664	0.1983	-0.0139	0.0896	-0.1914	0.1837	0.0097	0.0758	-0.0579	0.2037	0.0336	0.0863	-0.0499	0.1720
κ^Q	-0.0064	0.0744	-0.0606	0.2244	-0.0057	0.0923	-0.1718	0.2092	0.0163	0.0821	-0.0507	0.2337	0.0406	0.0920	-0.0456	0.1907
n quotes	24.0000	0.0000	24.0000	24.0000	24.0000	0.0000	24.0000	24.0000	23.3077	0.9473	21.0000	24.0000	24.0000	0.0000	24.0000	24.0000

Notes: This table reports Mean, SD, Min, and Max of the monthly Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), σ , $\lambda^* = \lambda$, μ_J (P), σ_J , κ (P), μ_J^* (Q), κ^Q , n quotes. Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P6 · GARCH-Merton · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ	-0.1720	0.0672	-0.2382	-0.0912	0.4934	0.2142	0.2381	0.8472	-0.1502	0.0471	-0.1991	-0.1082	0.3362	0.1151	0.2088	0.4477
ω	5.583e-06	1.525e-06	3.020e-06	8.070e-06	4.896e-05	1.467e-05	1.569e-05	6.376e-05	2.306e-05	8.714e-06	1.348e-05	4.182e-05	9.989e-05	8.167e-05	1.946e-05	2.546e-04
α	0.1112	0.0207	0.0584	0.1334	0.1195	0.0169	0.0957	0.1498	0.1625	0.0425	0.1033	0.2156	0.0600	0.0375	7.835e-04	0.1233
β	0.8788	0.0164	0.8539	0.9173	0.8368	0.0246	0.8079	0.8842	0.8189	0.0343	0.7719	0.8816	0.8609	0.0724	0.7193	0.9332
σ_0	0.0254	0.0126	0.0125	0.0501	0.0346	0.0189	0.0179	0.0859	0.0324	0.0126	0.0151	0.0544	0.0359	0.0072	0.0262	0.0536
λ	0.7675	0.2473	0.6632	1.3298	2.1505	0.4001	1.3333	2.6880	3.2274	0.7210	2.0000	4.0320	2.9168	0.6889	2.0000	3.9894
μ_J	-0.0774	0.0242	-0.0919	-0.0259	-0.1202	0.0177	-0.1438	-0.0851	0.0075	0.0133	-0.0138	0.0255	0.0640	0.0278	0.0296	0.1077
σ_J	0.0072	0.0175	0.0000	0.0467	0.0549	0.0143	0.0251	0.0759	0.1152	0.0178	0.0778	0.1282	0.1373	0.0108	0.1147	0.1599
κ	-0.0741	0.0229	-0.0878	-0.0256	-0.1117	0.0153	-0.1314	-0.0813	0.0145	0.0147	-0.0096	0.0340	0.0767	0.0304	0.0390	0.1256
R2 Normal																
μ	0.1945	2.889e-17	0.1945	0.1945	0.1781	0.1951	-0.0234	0.4142	0.1597	2.889e-17	0.1597	0.1597	0.1803	0.1245	0.0894	0.3665
ω	1.171e-05	2.976e-06	6.399e-06	1.438e-05	6.986e-06	2.680e-06	4.169e-06	1.143e-05	2.423e-05	1.253e-06	2.158e-05	2.584e-05	1.916e-04	1.036e-04	3.790e-05	2.962e-04
α	0.1609	0.0324	0.1289	0.2190	0.1112	0.0139	3.044e-09	0.0388	0.0052	0.0041	3.044e-09	0.0129	0.2888	0.0942	0.1244	0.4485
β	0.6094	0.0660	0.5096	0.7027	0.9635	0.0165	0.9333	0.9782	0.8832	0.0035	0.8758	0.8881	0.2058	0.3131	2.972e-09	0.7481
σ_0	0.0064	8.995e-04	0.0050	0.0084	0.0158	0.0023	0.0138	0.0216	0.0146	3.744e-04	0.0141	0.0151	0.0176	0.0037	0.0139	0.0293
λ	2.1485	0.7239	1.3194	3.3158	4.5527	0.4549	3.9894	5.3475	3.1216	0.3292	2.6387	3.3511	4.2428	1.0555	2.6596	5.3333
μ_J	-0.0192	0.0050	-0.0225	-0.0087	-0.0187	0.0219	-0.0541	0.0059	-0.0147	0.0034	-0.0169	-0.0095	-0.0114	0.0192	-0.0298	0.0427
σ_J	0.0062	0.0094	6.892e-04	0.0251	0.0705	0.0061	0.0636	0.0810	0.0835	0.0051	0.0802	0.0908	0.0902	0.0078	0.0670	0.1000
κ	-0.0190	0.0050	-0.0222	-0.0083	-0.0159	0.0212	-0.0505	0.0080	-0.0112	0.0038	-0.0136	-0.0054	-0.0071	0.0189	-0.0249	0.0460
R3 Late-cy																
μ	0.1419	2.889e-17	0.1419	0.1419	0.3715	0.1728	0.2471	0.6847	0.3944	0.0000	0.3944	0.3944	0.6509	0.1570	0.4450	0.9814
ω	3.669e-06	1.338e-06	1.902e-06	5.946e-06	2.255e-05	9.351e-06	1.112e-05	4.471e-05	2.908e-05	8.863e-06	1.543e-05	4.409e-05	3.378e-05	1.748e-05	1.364e-05	5.820e-05
α	0.1968	0.0232	0.1609	0.2312	0.1541	0.0243	0.1149	0.1837	0.2112	0.0305	0.1346	0.2628	0.1943	0.0322	0.1266	0.2361
β	0.7719	0.0201	0.7312	0.8059	0.7693	0.0579	0.6476	0.8554	0.6688	0.0706	0.5315	0.7973	0.7277	0.0505	0.6280	0.8170
σ_0	0.0095	0.0052	0.0043	0.0231	0.0168	0.0060	0.0108	0.0289	0.0154	0.0051	0.0096	0.0270	0.0203	0.0093	0.0126	0.0393
λ	3.3390	0.6051	2.6596	4.6299	4.3171	0.8081	3.3071	5.3760	2.9792	0.4371	2.0160	3.3600	1.5404	0.6852	0.6649	2.6737
μ_J	-0.0291	0.0022	-0.0315	-0.0241	-0.0174	0.0069	-0.0237	-0.0029	-0.0100	0.0142	-0.0431	0.0169	-0.0186	0.0383	-0.0635	0.0406
σ_J	0.0081	0.0023	0.0036	0.0107	0.0638	0.0074	0.0502	0.0705	0.0595	0.0143	0.0143	0.0741	0.0661	0.0539	0.0000	0.1327
κ	-0.0286	0.0022	-0.0310	-0.0238	-0.0152	0.0064	-0.0210	-0.0017	-0.0080	0.0144	-0.0421	0.0188	-0.0142	0.0412	-0.0615	0.0488
R4 COVID																
μ	0.2638	0.1528	0.0514	0.4889	0.6898	0.1796	0.5559	0.9935	0.3921	0.1664	0.2690	0.7066	0.3256	0.1042	0.2567	0.5552
ω	4.424e-06	6.550e-07	3.156e-06	5.375e-06	1.614e-05	2.458e-06	1.160e-05	1.932e-05	1.218e-05	3.260e-06	6.780e-06	1.976e-05	1.623e-05	5.488e-06	1.174e-05	2.927e-05
α	0.2421	0.0643	0.1543	0.3244	0.1660	0.0284	0.1352	0.2342	0.1831	0.0359	0.1346	0.2417	0.1490	0.0298	0.1031	0.1988
β	0.7368	0.0408	0.6756	0.7944	0.8055	0.0231	0.7523	0.8331	0.7793	0.0289	0.7276	0.8113	0.8108	0.0425	0.7184	0.8657
σ_0	0.0143	0.0128	0.0057	0.0494	0.0212	0.0118	0.0123	0.0542	0.0202	0.0147	0.0091	0.0612	0.0178	0.0062	0.0121	0.0335
λ	2.8227	0.6193	2.0000	4.0213	4.2074	1.1932	1.9895	5.3333	3.0284	0.7014	1.9895	4.0213	2.0013	0.6641	0.6667	3.3158
μ_J	-0.0314	0.0034	-0.0372	-0.0283	-0.0244	0.0108	-0.0442	-0.0023	-0.0288	0.0102	-0.0468	-0.0095	-0.0056	0.0342	-0.0635	0.0280
σ_J	0.0070	0.0052	0.0035	0.0160	0.0691	0.0068	0.0635	0.0885	0.0381	0.0129	0.0150	0.0613	0.0540	0.0275	0.0000	0.0799
κ	-0.0309	0.0032	-0.0364	-0.0279	-0.0217	0.0109	-0.0412	0.0016	-0.0275	0.0103	-0.0456	-0.0081	-0.0032	0.0347	-0.0615	0.0311

Notes: This table reports Mean, SD, Min, and Max of the monthly GARCH-Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , ω , α , β , σ_0 , λ , μ_J , σ_J , κ . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P7 · Heston-Merton · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

regime / para	SPY Mean	SD	Min	Max	AAPL Mean	SD	Min	Max	MSFT Mean	SD	Min	Max	AMZN Mean	SD	Min	Max
R1 Crisis	-0.2873 (P)	0.1250 (P)	-0.4424 (P)	-0.0608 (P)	-0.0499 (P)	0.2597 (P)	-0.2648 (P)	0.4609 (P)	-0.2249 (P)	0.1181 (P)	-0.3572 (P)	-0.0067 (P)	0.0101 (P)	0.2638 (P)	-0.3265 (P)	0.4808 (P)
R2 Normal	0.1779 (P)	0.0221 (P)	0.1476 (P)	0.2233 (P)	0.1615 (P)	0.2230 (P)	-0.1422 (P)	0.4632 (P)	0.2651 (P)	0.0662 (P)	0.1597 (P)	0.3537 (P)	0.1805 (P)	0.0801 (P)	0.0735 (P)	0.3665 (P)
R3 Late-cv	0.0701 (P)	0.0323 (P)	0.0222 (P)	0.1419 (P)	0.1466 (P)	0.0860 (P)	0.0279 (P)	0.3211 (P)	0.3024 (P)	0.0364 (P)	0.2586 (P)	0.3944 (P)	0.3128 (P)	0.1220 (P)	0.1069 (P)	0.5438 (P)
R4 COVID	0.0766 (P)	0.0617 (P)	0.0804 (P)	0.1512 (P)	0.3347 (P)	0.2027 (P)	0.0935 (P)	0.7360 (P)	0.3438 (P)	0.0843 (P)	0.2293 (P)	0.4846 (P)	0.1813 (P)	0.1760 (P)	0.0395 (P)	0.4934 (P)

Notes: This table reports Mean, SD, Min, and Max of the monthly Heston-Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), κ (P), θ (P), ξ , ρ , ν_0 (P), $\lambda^* = \lambda$, μ_J (P), σ_J , κ_J (P), η_{-v} (premium), κ^* (Q), θ^* (Q), $\nu_0^{\wedge}Q$, μ_J^* (Q), $\kappa_J^{\wedge}Q$, n quotes. Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.